

Leveraged S&P 500 Performance Analysis

Prepared by Harman Singh for Professor Phelim Boyle

June 2026

Data and Sample

The analysis uses monthly S&P 500 total return data from January 2017 to May 2026, for a total of 113 monthly observations. The data source is the Yahoo Finance S&P 500 Total Return Index series, ^SP500TR, which includes dividends and is therefore consistent with the paper's dividend-reinvestment convention. All statistics are reported monthly and are not annualised.

Methodology

Let r_t denote the S&P 500 total return in month t , in percent, and let r_f denote the constant monthly risk-free rate, also in percent. For each leverage level λ , the portfolio return is

$$R_t(\lambda) = \lambda r_t - (\lambda - 1)r_f = \lambda(r_t - r_f) + r_f.$$

The analysis uses $\lambda \in \{0.5, 1, 2, 3, 4, 5, 6\}$ and $r_f \in \{0.00\%, 0.25\%, 0.50\%\}$.

For each portfolio return series, the mean is

$$\bar{R} = \frac{1}{n} \sum_{t=1}^n R_t,$$

and the sample standard deviation is

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (R_t - \bar{R})^2}.$$

The Sharpe ratio is $\bar{R}/\sigma$. The Sortino ratio uses the zero-return threshold:

$$\sigma_d = \sqrt{\frac{1}{n} \sum_{t=1}^n \min(R_t, 0)^2}, \quad \text{Sortino} = \frac{\bar{R}}{\sigma_d}.$$

The Omega ratio is

$$\Omega = \frac{\sum_{t: R_t > 0} R_t}{\sum_{t: R_t < 0} |R_t|}.$$

Results

Table 1: Leveraged S&P 500 monthly statistics for $r_f = 0.00\%$.

λ	$\bar{R}$ (%)	σ (%)	Sharpe	Sortino	Omega
0.5	0.6638	2.2719	0.2922	0.4734	2.0770
1	1.3275	4.5437	0.2922	0.4734	2.0770
2	2.6551	9.0875	0.2922	0.4734	2.0770
3	3.9826	13.6312	0.2922	0.4734	2.0770
4	5.3102	18.1750	0.2922	0.4734	2.0770
5	6.6377	22.7187	0.2922	0.4734	2.0770
6	7.9653	27.2624	0.2922	0.4734	2.0770

Table 2: Leveraged S&P 500 monthly statistics for $r_f = 0.25\%$.

λ	$\bar{R}$ (%)	σ (%)	Sharpe	Sortino	Omega
0.5	0.7888	2.2719	0.3472	0.5852	2.3582
1	1.3275	4.5437	0.2922	0.4734	2.0770
2	2.4051	9.0875	0.2647	0.4205	1.9470
3	3.4826	13.6312	0.2555	0.4033	1.9050
4	4.5602	18.1750	0.2509	0.3948	1.8843
5	5.6377	22.7187	0.2482	0.3897	1.8719
6	6.7153	27.2624	0.2463	0.3863	1.8637

Table 3: Leveraged S&P 500 monthly statistics for $r_f = 0.50\%$.

λ	$\bar{R}$ (%)	σ (%)	Sharpe	Sortino	Omega
0.5	0.9138	2.2719	0.4022	0.7058	2.6755
1	1.3275	4.5437	0.2922	0.4734	2.0770
2	2.1551	9.0875	0.2372	0.3696	1.8231
3	2.9826	13.6312	0.2188	0.3367	1.7438
4	3.8102	18.1750	0.2096	0.3205	1.7052
5	4.6377	22.7187	0.2041	0.3109	1.6824
6	5.4653	27.2624	0.2005	0.3045	1.6673

Brief Discussion

When $r_f = 0.00\%$, leverage scales both average return and volatility almost proportionally. Sharpe, Sortino, and Omega are unchanged across leverage levels because every portfolio return is a constant multiple of the same underlying S&P 500 total return series.

When $r_f = 0.25\%$, borrowing costs reduce the return earned at higher leverage levels. The mean return still rises with λ , but Sharpe, Sortino, and Omega decline as leverage increases because the cost of financing lowers returns without reducing downside exposure.

When $r_f = 0.50\%$, the effect of borrowing costs is stronger. The higher-leverage portfolios still have larger mean returns, but the risk-adjusted ratios fall more sharply, showing that leverage is less attractive when the monthly financing cost is higher.